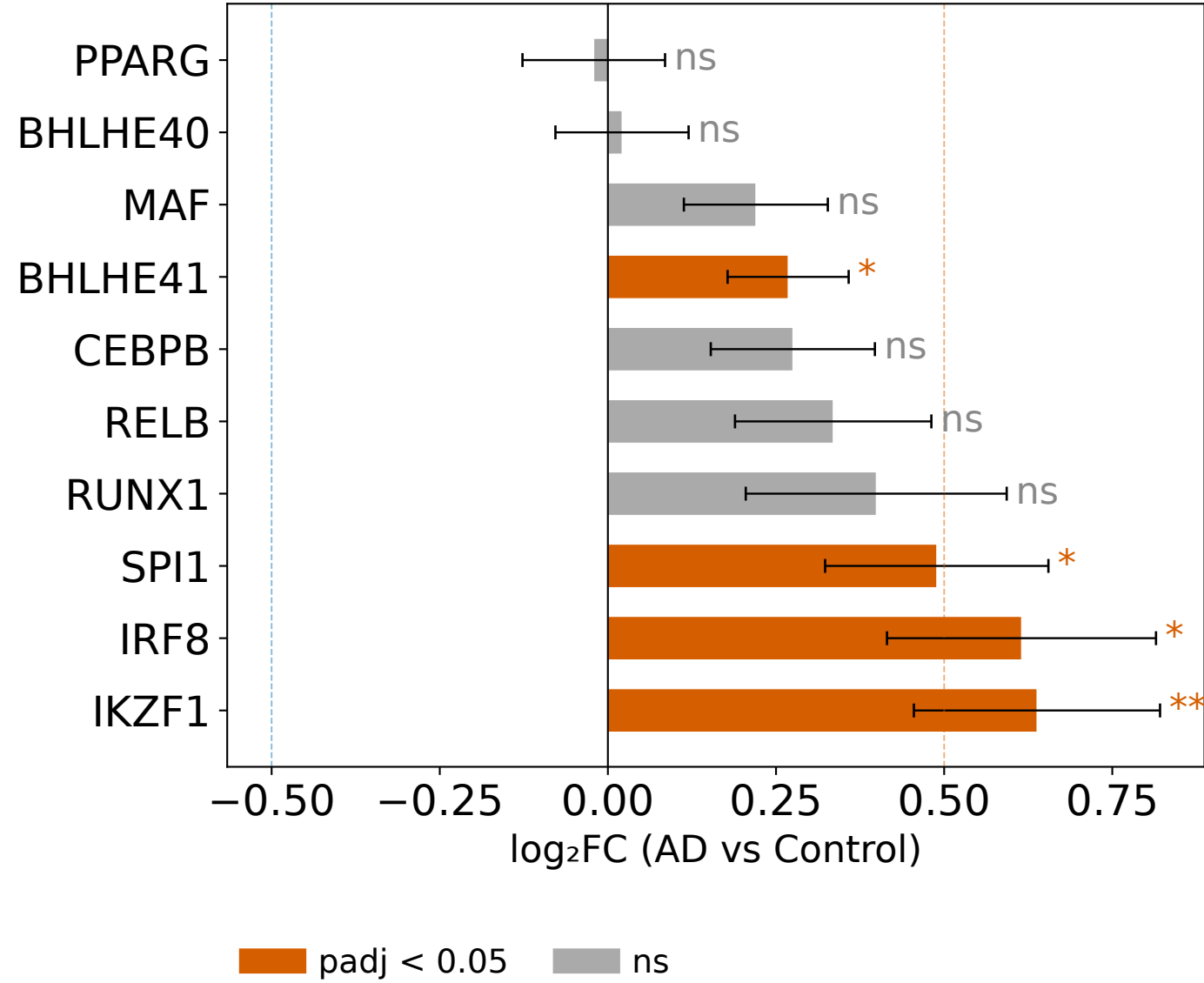
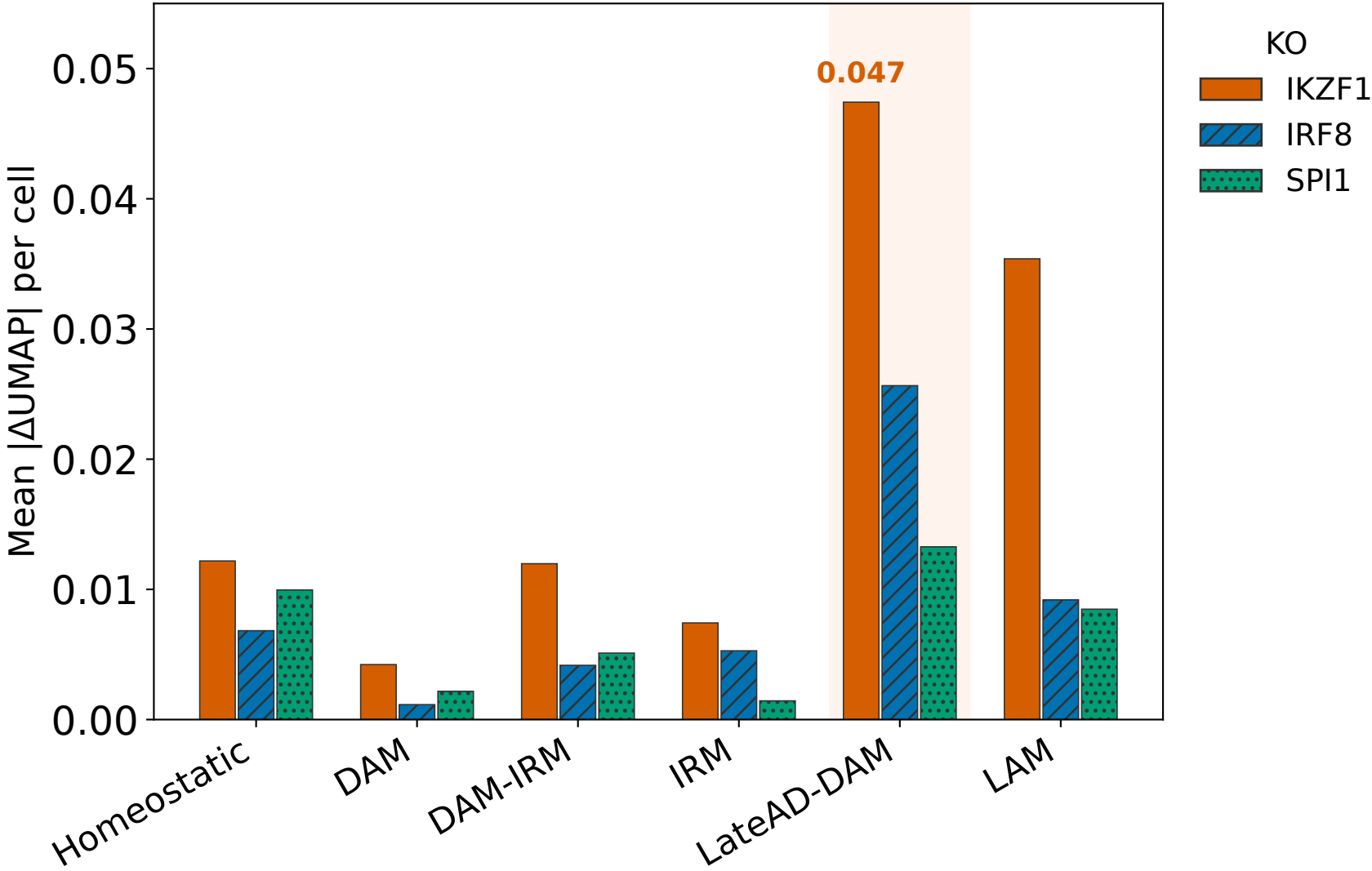


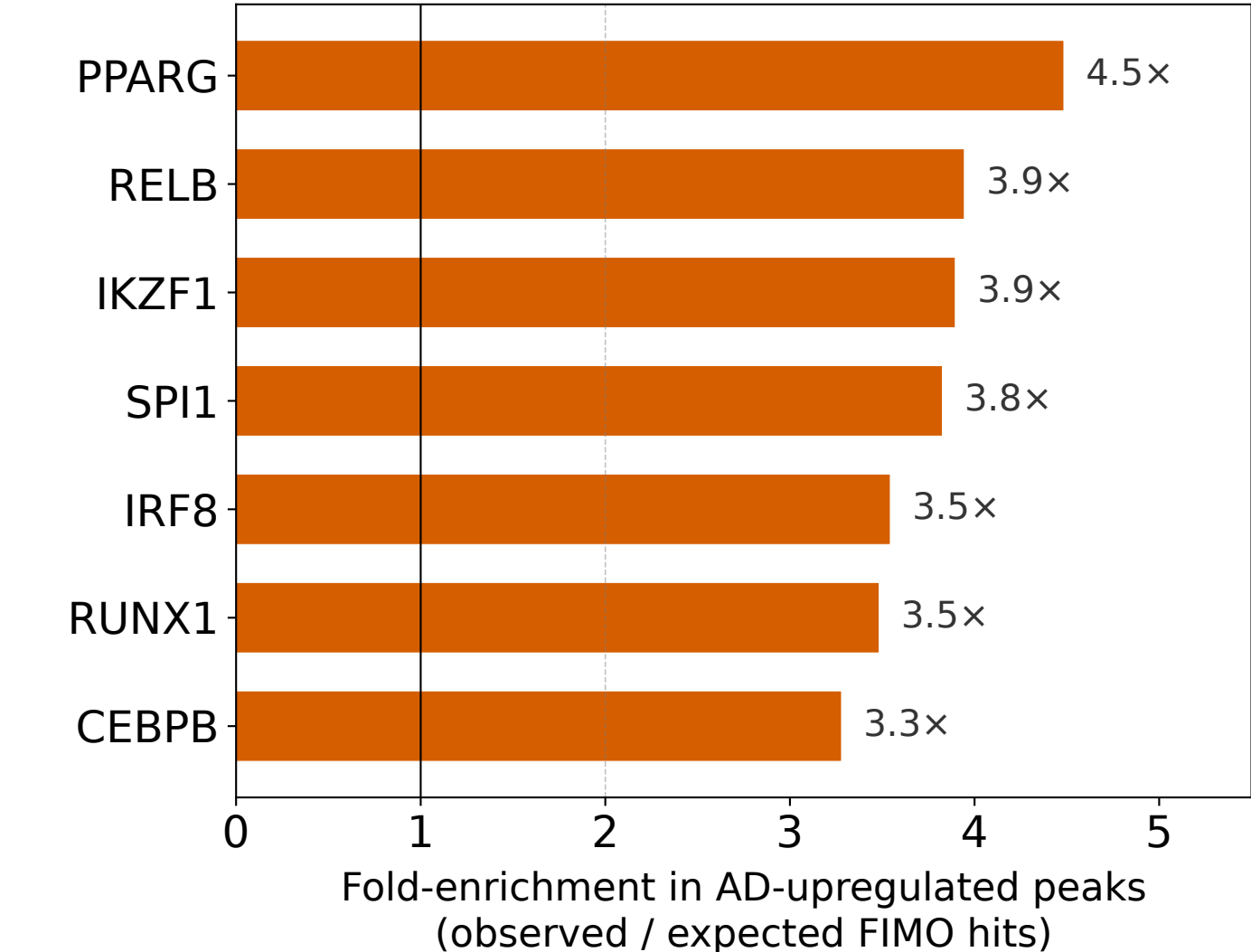
A GSE95587 bulk replication
(fusiform gyrus, n=117)



B CellOracle in-silico KO perturbation
(UMAP shift magnitude)



C ATAC chromatin accessibility
(motif enrichment in 56 AD-up peaks, n=1,191 DAPs)



D IKZF1 convergent validation
(6 independent lines of evidence)

Modality	Method	Effect Size	Significance
Differential expression	DESeq2 pseudobulk	log ₂ FC = +1.4	padj < 0.001
Regulon activity (AUCell)	pySCENIC (5-seed)	AUC = 0.153	peak: LateAD-DAM
Pseudotime correlation	Spearman	ρ = +0.309	p < 0.001
Bulk replication	DESeq2 (GSE95587)	log ₂ FC = +0.638	padj = 0.004
Chromatin accessibility	FIMO motif enrichment	3.9× in AD-up	56/1,191 peaks
In-silico KO	CellOracle GRN	Δmag = 0.047	largest perturbation

* log₂FC (DGE) from internal pseudobulk DESeq2, DAM vs Homeostatic.